

Wave Basics + Standard Form

Answer sheet · 26 marks total

Section A — Label the diagram

A = peak (crest) B = amplitude C = wavelength D = trough

[1 mark each, 4 marks total]

Section B — Wave concepts

B1. The particles vibrate at right angles to the direction the wave travels (1). This is unlike a longitudinal wave, where particles move along the same line as the wave (1).

B2. Sound (or any valid example, e.g. a pushed slinky/spring). (1)

B3. $v = f \times \lambda$ (1) $v = 25 \times 12$ (1) $v = 300 \text{ m/s}$ (1)

B4. Rearrange $v = f\lambda \rightarrow \lambda = v \div f$ (1) $\lambda = 320 \div 40$ (1) $\lambda = 8 \text{ m}$ (1)

B5. The wave with the larger amplitude carries more energy (1), because amplitude is linked to how much energy the wave transfers (1).

Section C — Standard form practice

C1. 6.2×10^3

C2. 3.4×10^{-3}

C3. 25 000

C4. 0.007

C5. $1 \times 10^{-11} \text{ m}$

C6. $1.2 \times 10^3 \text{ m}$

C7. $3 \times 10^8 \text{ m/s}$

C8. $2.5 \times 10^{-4} \text{ s}$

C9. 0.000056

C10. $4.5 \times 10^{-4} < 3 \times 10^{-2} < 2 \times 10^{-1}$ (1 mark for correct order, 1 mark for correct reasoning/working shown)

Marking note: for calculation questions, award partial credit for correct method even if the final answer is wrong (e.g. correct equation and substitution, but an arithmetic slip at the end).